



**SWAMINARAYAN
UNIVERSITY**
THE RESERVOIR OF ALL KNOWLEDGE

Faculty of Engineering (Degree)
Semester – 5th
Subject Name –Computer Networks
Subject Code - BE103404
Faculty Name – Radhika Gokani
Chapter -3

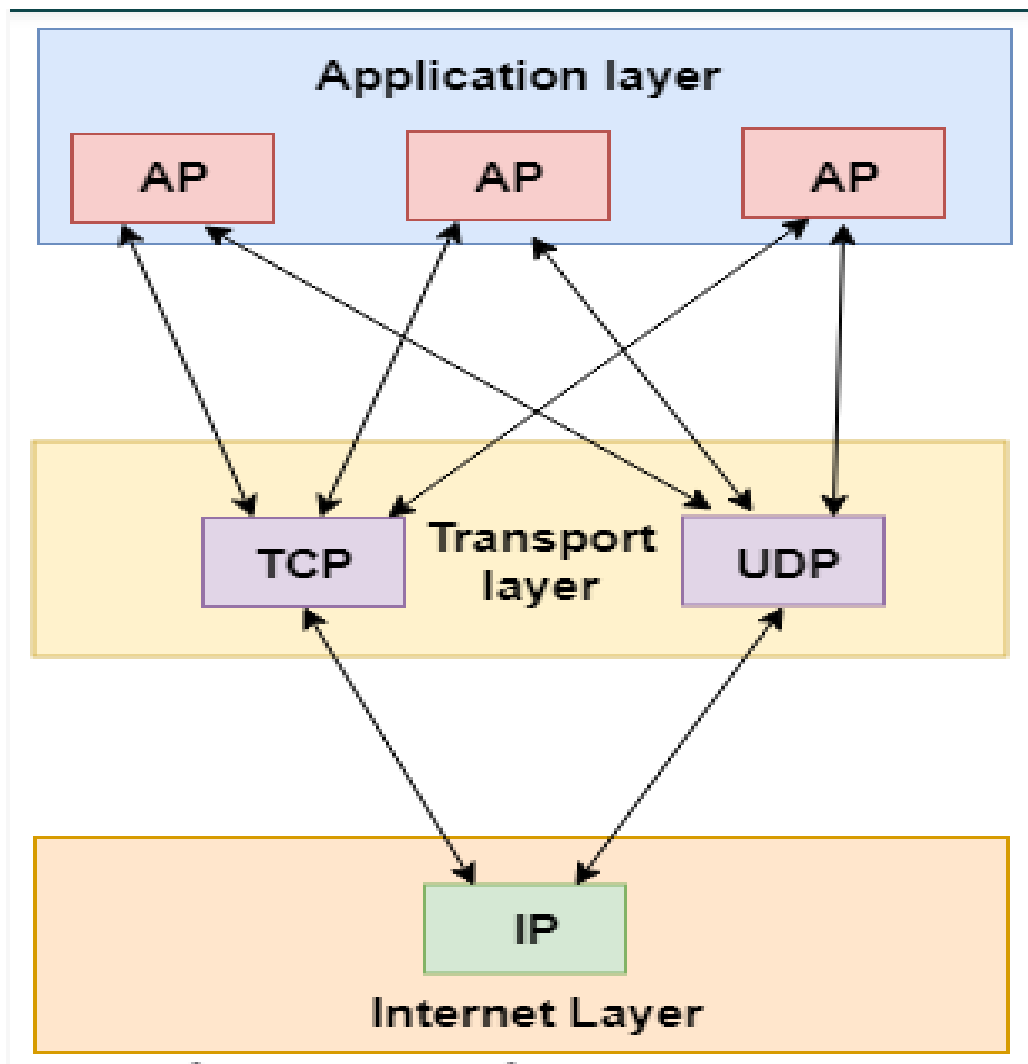
Transport Layer

3 & 3.1 introduction and Transport layer services

The **transport layer** is a crucial component of computer networks, specifically in the OSI (Open Systems Interconnection) model, where it is the fourth layer

Functions of the Transport Layer

1. **End-to-End Communication:** The transport layer ensures reliable data transfer between host systems. It establishes a connection, maintains it during data transfer, and terminates it once the communication is complete.
2. **Segmentation and Reassembly:** It breaks down large data from the application layer into smaller segments for transmission. At the destination, these segments are reassembled into the original data stream.
3. **Flow Control:** This function prevents network congestion by controlling the rate of data transmission. Protocols like the sliding window protocol are used to manage this.
4. **Error Detection and Correction:** The transport layer uses checksums to detect errors in the transmitted data. If errors are found, it requests retransmission of the corrupted segments.
5. **Multiplexing and Demultiplexing:** It allows multiple applications to use the network simultaneously by assigning unique identifiers to each data stream, ensuring that data is delivered to the correct application.



Common Transport Layer Protocols

- **TCP (Transmission Control Protocol):** Provides reliable, ordered, and error-checked delivery of data.

The transport layer plays a vital role in ensuring that data is transmitted accurately and efficiently across networks, making it an essential part of modern communication systems.

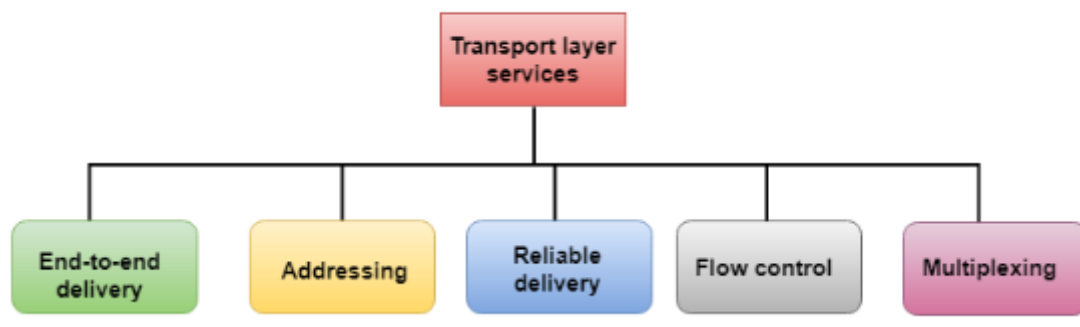
- **UDP (User Datagram Protocol):** Offers a simpler, connectionless service with minimal error recovery, suitable for applications needing fast, efficient transmission
 - The transport layer is a 4th layer from the top.
 - The main role of the transport layer is to provide the communication services directly to the application processes running on different hosts.
 - The transport layer provides a logical communication between application processes running on different hosts. Although the application processes on

different hosts are not physically connected, application processes use the logical communication provided by the transport layer to send the messages to each other.

- The transport layer protocols are implemented in the end systems but not in the network routers.
- A computer network provides more than one protocol to the network applications. For example, TCP and UDP are two transport layer protocols that provide a different set of services to the network layer.
- All transport layer protocols provide multiplexing/demultiplexing service. It also provides other services such as reliable data transfer, bandwidth guarantees, and delay guarantees.
- Each of the applications in the application layer has the ability to send a message by using TCP or UDP. The application communicates by using either of these two protocols. Both TCP and UDP will then communicate with the internet protocol in the internet layer. The applications can read and write to the transport layer. Therefore, we can say that communication is a two-way process.

Services provided by the Transport Layer

The services provided by the transport layer are similar to those of the data link layer. The data link layer provides the services within a single network while the transport layer provides the services across an internetwork made up of many networks. The data link layer controls the physical layer while the transport layer controls all the lower layers.



The services provided by the transport layer protocols can be divided into five categories:

- End-to-end delivery
 - Addressing
 - Reliable delivery
 - Flow control
 - Multiplexing
-

3.2 Multiplexing and Demultiplexing

Multiplexing and **demultiplexing** are essential processes in computer networks, particularly at the transport layer.

Multiplexing/demultiplexing

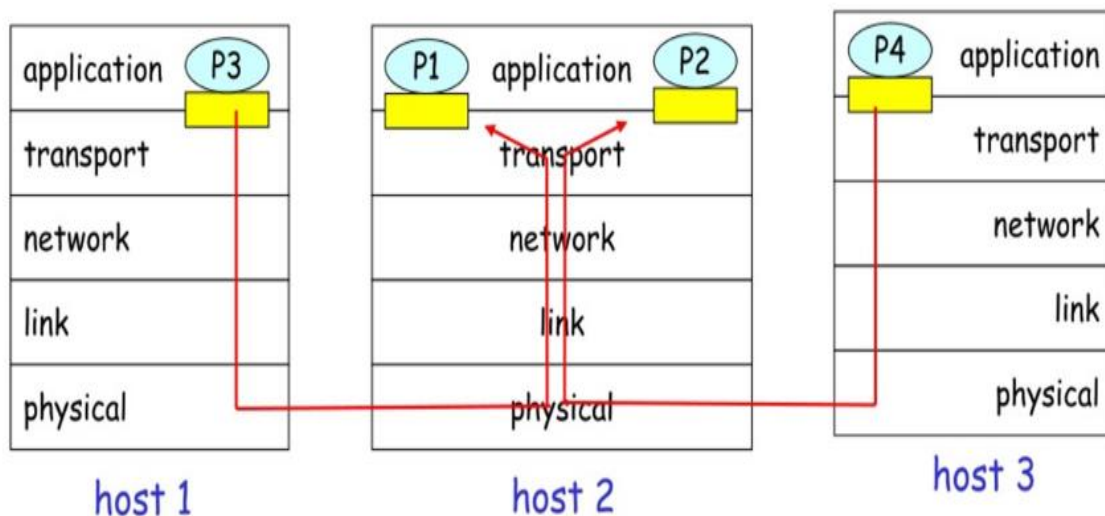
Demultiplexing at rcv host:

delivering received segments to correct socket

Multiplexing at send host:

gathering data from multiple sockets, enveloping data with header (later used for demultiplexing)

■ = socket ○ = process



Multiplexing

Multiplexing involves combining multiple data streams from different applications or sources into a single, unified data stream for transmission over a network. This process is crucial for efficient use of network resources. There are several types of multiplexing:

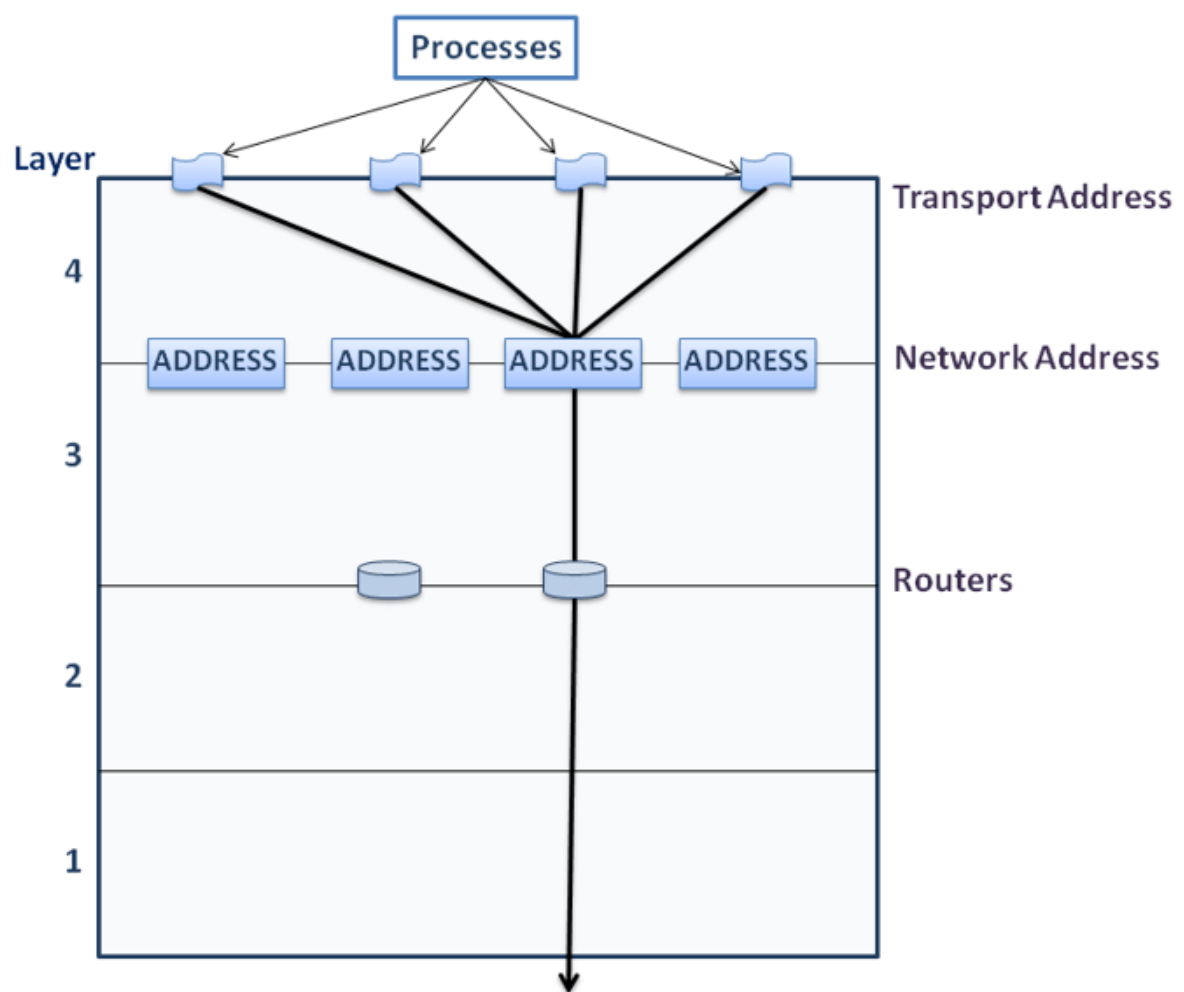
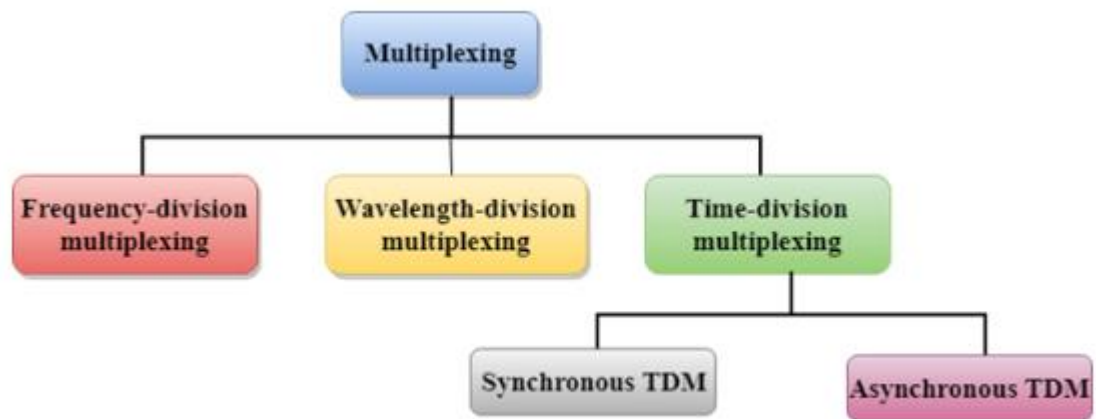


Figure 1: Multiplexing Mechanism

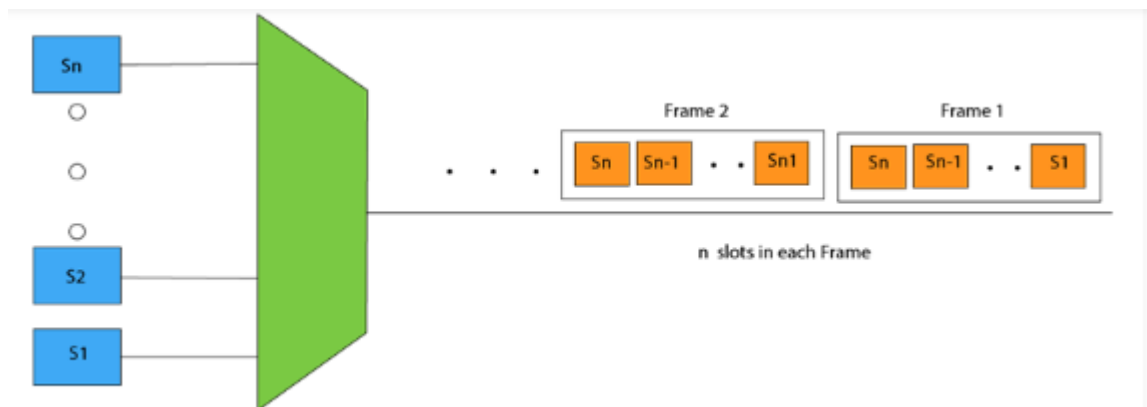


1. **Time-Division Multiplexing (TDM):** Divides the available bandwidth into time slots, with each data stream assigned a specific time slot.

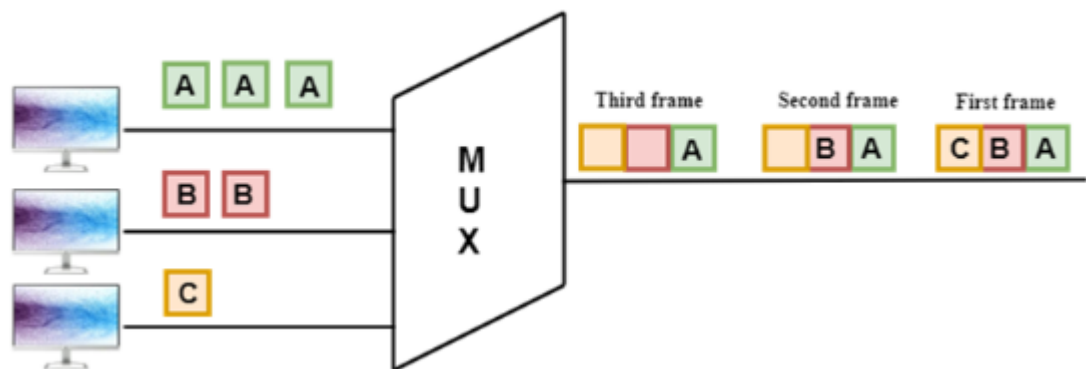
- It is a digital technique.
- In Frequency Division Multiplexing Technique, all signals operate at the same time with different frequency, but in case of Time Division Multiplexing technique, all signals operate at the same frequency with different time.
- In **Time Division Multiplexing technique**, the total time available in the channel is distributed among different users. Therefore, each user is allocated with different time interval known as a Time slot at which data is to be transmitted by the sender.
- A user takes control of the channel for a fixed amount of time.
- In Time Division Multiplexing technique, data is not transmitted simultaneously rather the data is transmitted one-by-one.
- In TDM, the signal is transmitted in the form of frames. Frames contain a cycle of time slots in which each frame contains one or more slots dedicated to each user.
- It can be used to multiplex both digital and analog signals but mainly used to multiplex digital signals.

There are two types of TDM:

- **Synchronous TDM**

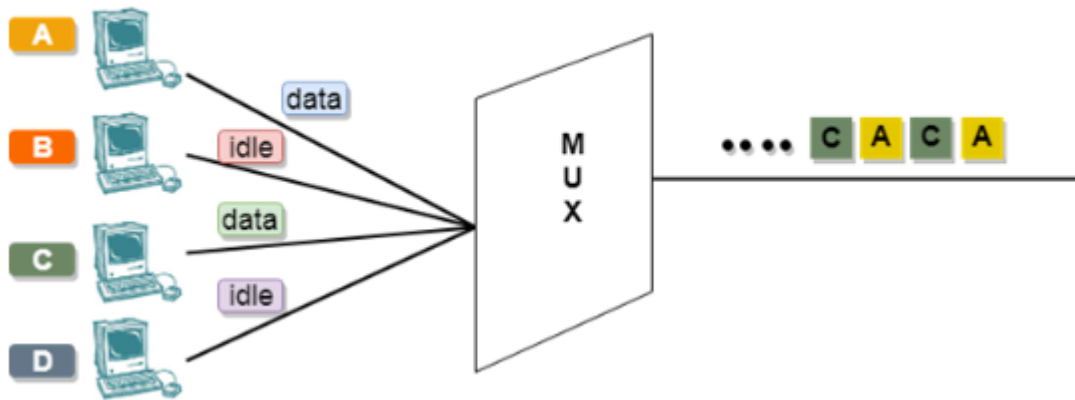


Concept Of Synchronous TDM

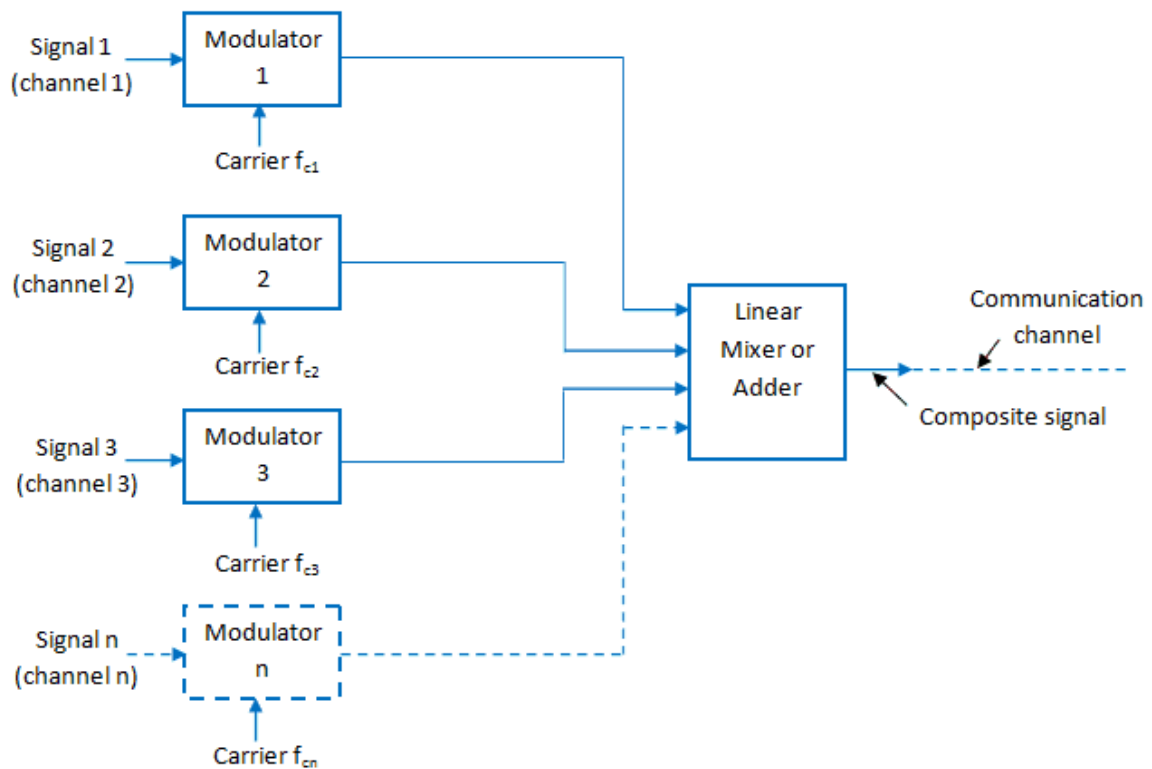


Asynchronous TDM

Concept Of Asynchronous TDM



2. **Frequency-Division Multiplexing (FDM):** Allocates different frequency bands to different data streams.



Frequency-Division Multiplexing

Frequency-Division Multiplexing (FDM) is a technique used in telecommunications to transmit multiple signals simultaneously over a single communication medium. Here's a brief overview:

How FDM Works

- **Bandwidth Division:** The total bandwidth of the communication medium is divided into a series of non-overlapping frequency bands. Each band carries a separate signal.
- **Carrier Signals:** Each signal is modulated onto a different carrier frequency. This means that each signal is transmitted at a unique frequency, preventing interference between signals.
- **Multiplexing and Demultiplexing:** At the transmitting end, signals are combined (multiplexed) and sent over the medium. At the receiving end, they are separated (demultiplexed) back into their original signals.

Applications of FDM

- **Radio and TV Broadcasting:** Multiple radio and TV channels are transmitted simultaneously over the airwaves.
- **Cable Television:** Many TV channels are carried on a single cable.
- **Telephone Systems:** Multiple telephone calls are transmitted through high-capacity trunk lines.
- **DSL Modems:** Large amounts of computer data are transmitted through twisted pair telephone lines.

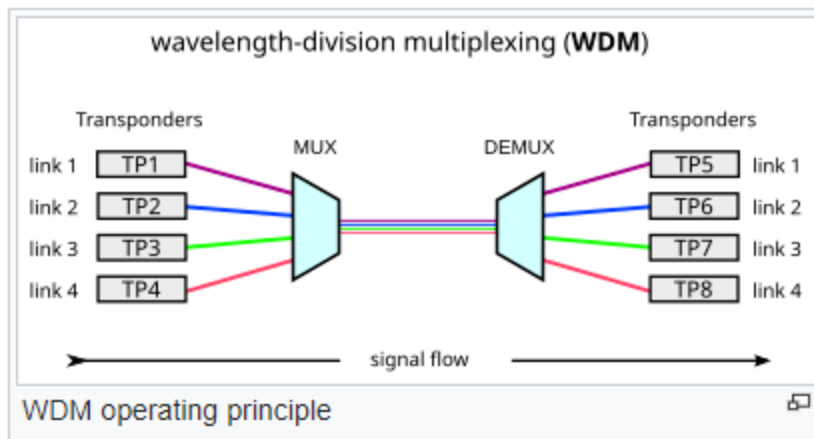
Advantages

- **Efficient Use of Bandwidth:** Multiple signals can be transmitted over a single channel.
- **No Time Synchronization Required:** Easier to implement as it doesn't require precise time synchronization.
- **Low Implementation Cost:** Relatively simple and inexpensive to implement.

Disadvantages

- **Limited Capacity:** The number of signals that can be transmitted is limited by the available bandwidth.

-
-
3. **Wavelength-Division Multiplexing (WDM):** Used in fiber-optic communications, where different wavelengths (colors) of light carry different data streams.



Wavelength-Division Multiplexing (WDM) is a technology used in fiber-optic communications to increase bandwidth by allowing multiple data streams to be transmitted simultaneously over a single optical fiber. This is achieved by using different wavelengths (or colors) of laser light for each data stream.

There are two main types of WDM:

1. **Coarse Wavelength-Division Multiplexing (CWDM):** This type uses fewer wavelengths (typically less than 8) and is suitable for short-range communications. It is cost-effective but less efficient in terms of spectral usage.

2. **Dense Wavelength-Division Multiplexing (DWDM):** This type uses many more wavelengths (often 40 or more) and is used for long-distance and high-capacity applications. It is more expensive but allows for much higher data transmission rates.

WDM systems use a multiplexer at the transmitter to combine the different wavelengths and a demultiplexer at the receiver to separate them. This technology allows telecommunications companies to expand their network capacity without laying additional fiber.

Demultiplexing

Demultiplexing is the reverse process of multiplexing. At the receiving end, the combined data stream is separated back into its original individual data streams and delivered to the appropriate applications. This ensures that data is correctly routed to its intended destination.

Demultiplexing in computer networking is a crucial process that occurs at the transport layer of the OSI model. It involves directing incoming data packets to the correct application process based on information in the packet headers, such as port numbers.

1. **Transport Layer Role:** The transport layer protocols, such as TCP and UDP, handle demultiplexing. They use port numbers to identify the receiving application. For instance, TCP relies on a combination of IP addresses and port numbers, while UDP uses only port numbers.
2. **Process:** When data packets arrive at the destination, the transport layer examines the destination port number in the packet header. This port number helps the transport layer determine which application should receive the data. The process of separating these packets and delivering them to the appropriate application is called demultiplexing.
3. **Types of Demultiplexing:**
 - **Connectionless Demultiplexing:** Used by UDP, where each packet is treated independently, and the destination port number is used to direct the packet to the correct application.
 - **Connection-Oriented Demultiplexing:** Used by TCP, where a connection is established between the sender and receiver before data transfer. The connection is identified by a combination of source and destination IP addresses and port numbers.
4. **Example:** Imagine you are using multiple applications on your device, like a web browser and an email client. When data packets arrive, demultiplexing ensures that web data goes to the browser and email data goes to the email client, even though both types of data are coming through the same network connection.

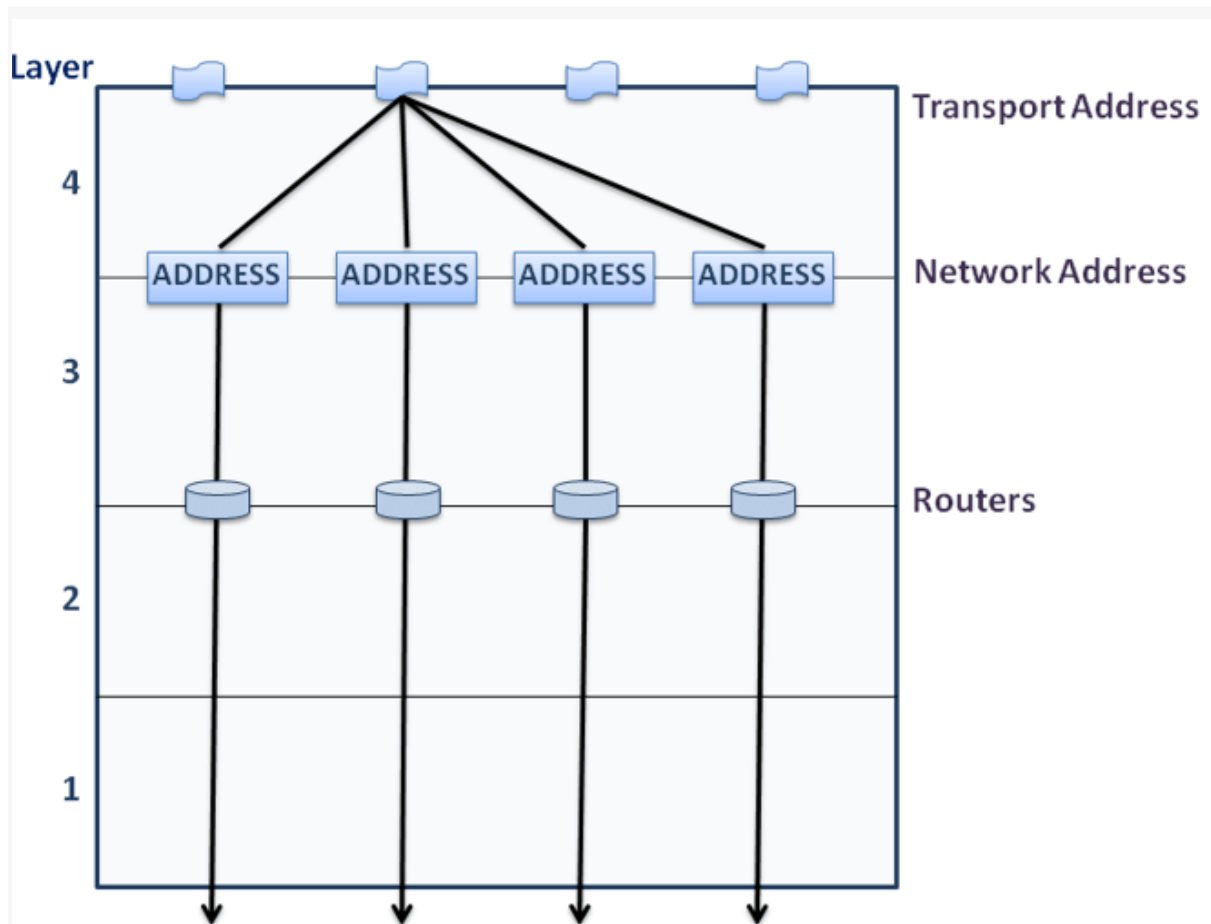


Figure 2: De-Multiplexing Mechanism



How It Works in the Transport Layer

In the transport layer, protocols like TCP and UDP handle multiplexing and demultiplexing using port numbers:

- **Multiplexing:** The transport layer collects data from multiple applications, adds a header containing source and destination port numbers, and sends it as a single stream.
- **Demultiplexing:** At the receiving end, the transport layer examines the port numbers in the header to determine which application should receive the data.

Practical Example

Imagine you are using multiple applications on your computer, such as a web browser and an email client. When data is sent from these applications, multiplexing combines the data into a single stream. Upon reaching the destination, demultiplexing ensures that the web browser data goes to the web browser and the email data goes to the email client.

These processes are fundamental for efficient and accurate data transmission in modern networks.

3.3 Connectionless transport (UDP)

The **User Datagram Protocol (UDP)** is a connectionless transport layer protocol used in computer networks. Unlike the Transmission Control Protocol (TCP), UDP does not establish a connection before sending data, making it faster but less reliable. Here are some key points about UDP:

1. **Connectionless Nature:** UDP sends data packets, called datagrams, without establishing a connection between the sender and receiver. Each packet is sent independently, and they may take different paths to reach the destination.
2. **No Guarantee of Delivery:** UDP does not guarantee the delivery, order, or integrity of the data packets. This means packets can be lost, duplicated, or received out of order.
3. **Low Overhead:** The protocol has minimal overhead, with a simple header structure of only 8 bytes, making it efficient for applications that require fast transmission and can tolerate some data loss.
4. **Applications:** UDP is commonly used in applications where speed is critical and occasional data loss is acceptable.

Examples include:

- **Streaming media** (audio and video)
- **Online gaming**
- **Voice over IP (VoIP)**
- **Domain Name System (DNS) queries**

5. **Port Numbers:** UDP uses port numbers to distinguish different user requests or processes. Each datagram includes source and destination port numbers

3.4 Principles of reliable data transfer

Reliable data transfer in computer networks ensures that data is transmitted accurately and in the correct order, despite potential issues like data loss, corruption, or duplication.

Principles:

1. **Error Detection and Correction:** Mechanisms like checksums and cyclic redundancy checks (CRC) are used to detect errors in transmitted data. If errors are detected, protocols like TCP can request retransmission of the corrupted data.
2. **Acknowledgments (ACKs) and Negative Acknowledgments (NAKs):** The receiver sends an ACK to the sender to confirm successful receipt of data. If data is corrupted or lost, a NAK is sent, prompting the sender to retransmit the data.
3. **Sequence Numbers:** Each packet of data is assigned a unique sequence number. This helps in reordering packets that may arrive out of order and in detecting duplicate packets.
4. **Flow Control:** This ensures that the sender does not overwhelm the receiver with too much data at once. Protocols like TCP use a sliding window mechanism to manage the flow of data.
5. **Congestion Control:** This prevents network congestion by adjusting the rate of data transmission based on network conditions. TCP uses algorithms like slow start and congestion avoidance to manage congestion.
6. **Timeouts and Retransmissions:** If an acknowledgment is not received within a certain time frame, the sender assumes the packet was lost and retransmits it.

These principles are implemented in various layers of the network protocol stack, with TCP being a prime example of a protocol that ensures reliable data transfer over an unreliable network layer.

3.5 Connection-oriented transport (TCP)

Transmission Control Protocol (TCP) is a fundamental protocol in computer networks, specifically at the transport layer of the OSI model. Here are some key points about TCP:

1. **Connection-Oriented:** TCP is a connection-oriented protocol, meaning it establishes a connection between the sender and receiver before data transmission begins. This ensures that data packets are delivered in the correct order and without errors.
2. **Reliable Data Transfer:** TCP provides reliable data transfer by using acknowledgments and retransmissions. If a packet is lost or corrupted during transmission, TCP will retransmit it.
3. **Flow Control:** TCP uses flow control mechanisms to prevent network congestion. It adjusts the rate of data transmission based on the receiver's ability to process the incoming data.
4. **Error Detection and Correction:** TCP includes error-checking mechanisms to detect any errors in the transmitted data. If errors are found, the data is retransmitted.
5. **Segmentation and Reassembly:** TCP breaks down large data messages into smaller segments for transmission. These segments are reassembled into the original message at the receiving end.
6. **Applications:** TCP is widely used in applications where reliable data transfer is crucial, such as web browsing (HTTP/HTTPS), email (SMTP), and file transfer (FTP)

Types of Connection-Oriented Service :

Service	Example
Reliable Message Stream	Sequence of pages, etc.
Reliable Byte Stream	Song Download, etc.
Unreliable Connection	VoIP (Voice Over Internet Protocol)

3.6 Congestion control

Congestion control in computer networks is essential for maintaining smooth and efficient data flow. When too much data is sent through the network at once, it can lead to delays and data loss, similar to traffic congestion on a busy road. Here are some key points about congestion control:

What is Congestion?

Congestion occurs when the network is overloaded with data, causing delays and potential data loss. This can significantly impact the performance and reliability of the network.

Effects of Congestion Control

1. **Improved Network Stability:** Helps prevent network overload, ensuring stable and reliable performance.
2. **Reduced Latency and Packet Loss:** Manages traffic to minimize delays and data loss.
3. **Enhanced Throughput:** Optimizes resource usage, allowing more data to be transmitted efficiently.
4. **Fairness in Resource Allocation:** Ensures equitable bandwidth distribution among users.
5. **Better User Experience:** Smooth data flow leads to faster and more reliable applications and services.

6. **Mitigation of Network Congestion Collapse:** Prevents severe congestion that can make the network almost unusable.

Congestion Control Algorithms

1. **Leaky Bucket Algorithm:** Used for traffic shaping or rate-limiting, ensuring data is sent at a consistent rate.
2. **Token Bucket Algorithm:** Allows bursty data transmission while controlling the average data rate.

Congestion Control Techniques

1. **Open Loop Congestion Control:** Prevents congestion before it happens through policies like retransmission, window, discarding, acknowledgment, and admission policies.
2. **Closed Loop Congestion Control:** Addresses congestion after it occurs using techniques like backpressure, where a congested node stops receiving packets from upstream nodes

3.7 TCP congestion control

TCP (Transmission Control Protocol) congestion control is a critical mechanism used to manage data flow and prevent network congestion. Here's a brief overview of how it works:

Phases of TCP Congestion Control

1. **Slow Start Phase:**
 - **Exponential Growth:** Initially, the congestion window (CWND) size increases exponentially. For example, if the initial CWND is 1 segment, it doubles after each round-trip time (RTT) as long as acknowledgments are received successfully.
2. **Congestion Avoidance Phase:**
 - **Additive Increase:** Once the CWND reaches a threshold (ssthresh), it grows linearly. For instance, if the CWND is 20 segments, it increases by 1 segment per RTT.
3. **Congestion Detection Phase:**
 - **Multiplicative Decrease:** When congestion is detected (e.g., through packet loss or timeout), the CWND is reduced. If a timeout occurs, the CWND is set to 1, and the ssthresh is halved. If three duplicate ACKs are received, the CWND is set to the ssthresh value.

Algorithms and Techniques

- **Additive Increase/Multiplicative Decrease (AIMD):** This algorithm combines linear growth of the CWND with an exponential reduction when congestion occurs, ensuring fair bandwidth distribution among multiple flows.

- **Congestion Window (CWND):** This is a dynamic value maintained by the sender to control the number of unacknowledged packets in transit, preventing network overload.

Importance

TCP congestion control ensures efficient and stable data transmission across networks, adapting to current network conditions to avoid congestion collapse and maintain reliable communication